

SMART INDIA HACKATHON 2026

TRUETRACK



SMART INDIA
HACKATHON
2026

- Problem Statement ID – SIH168
- Problem Statement Title – Intelligent Dead Reckoning (IDR) System with GNSS Fusion using AI/ML
- Theme – Transportation & Logistics
- PS Category – Software
- Team ID – —
- Team Name – TrueTrack



❖ Proposed Solution

The Problem We Solve:

- GNSS blind spots: tunnels, underground metros, multi-level parking, urban canyons.
- Millions of two-wheelers, **trucks & older cars** have **no OBD-II or factory INS fallback**.
- Low-cost dashboard IMUs are noisy — vibration & mount drift make raw DR drift within seconds.
- Result: apps freeze/jump, causing missed exits, delivery delays and safety risk.

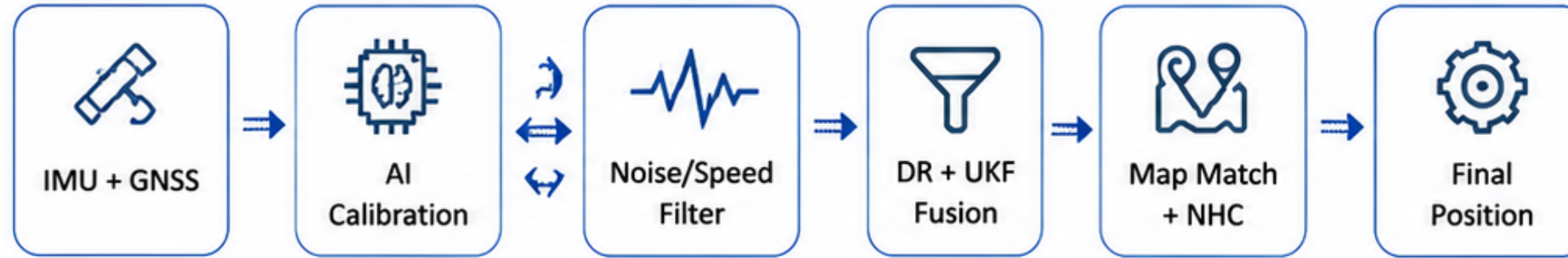
Our AI-Fused Dead Reckoning Solution:

1. **AI In-Vehicle Calibration** Auto-aligns phone pitch/roll/yaw to driving direction — works on any mount.
2. **AI Noise & Speed Correction** 1D-CNN/GRU filters IMU vibration and predicts vehicle speed, no OBD needed.
3. **UKF GNSS/INS Fusion** Fuses corrected IMU with GNSS; switches to inertial-only within ms of blackout.
4. **Offline Map Matching + NHC** Snaps trajectory back onto the real road using offline OSM road graph, within seconds.
5. Result: apps freeze/jump, causing missed exits, delivery delays and safety risk.

Innovation & Uniqueness: Unlike OBD/wheel-tick-dependent INS, TrueTrack works purely Off phone sensors on any vehicle, with automatic mount calibration, millisecond GNSS/INS handoff, and one trained model exported to both a mobile app and an edge engine.



TECHNICAL APPROACH



- **Model storage:** quantized 1D-CNN+GRU exported to TFLite/ONNX (~2–5 MB on-device); full-precision PyTorch checkpoints versioned server-side.
- **Database:** on-device Room/SQLite for trip logs + offline OSM tiles (MBTiles); backend PostgreSQL/PostGIS for trajectories & model versions.
- **Key features:** IMU (accel/gyro/mag), GNSS (lat, lon, speed, heading, HDOP), engineered features (jerk, vibration RMS, turn rate).
- **Innovation:** AI unified smartphone-only Dead Reckoning + GNSS fusion system combining AI noise correction, UKF fusion, offline map matching, and a dual app + edge-engine deployment for real-time, drift-free navigation.

TECH STACK

Mobile Application

- Kotlin / Android
- Sensor & Fused Location APIs

AI/ML

- PyTorch (train) ; ONNX/TFLite (deploy)
- 1D-CNN + GRU sequence model

Navigation & Fusion

- Unscented Kalman Filter
- Non-Holonomic Constraints

Maps & GIS

- Offline OSM extracts
- HMM map matching, MapLibre

Backend

- Python, FastAPI
- PostgreSQL + PostGIS

Edge Deployment

- ONNX Runtime edge engine
- External / FOG-grade IMU interface



SMART INDIA
HACKATHON
2026



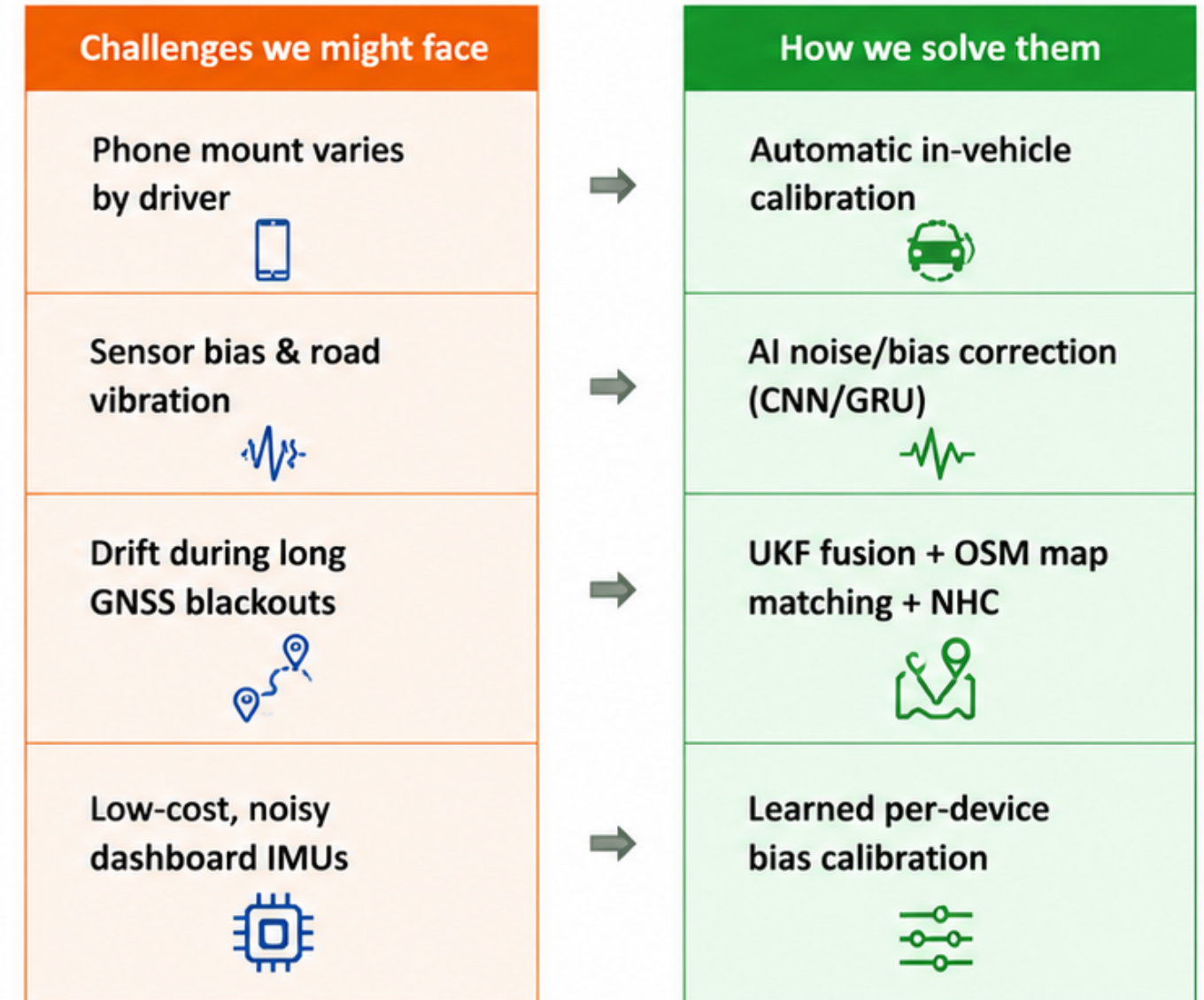
FEASIBILITY AND VIABILITY

I. FEASIBILITY: The Plan Is Actionable

- **Technical Feasibility**
 - 1D-CNN is a public, pre-synchronized dataset — training can start immediately.
 - Kalman/UKF fusion and map matching are mature, proven INS techniques.
- **Implementation Feasibility**
 - Every pipeline stage (baseline — AI correction — blackout test — map matching) is independently demonstrable.
 - Deliverable matches the PS exactly: one model exported to app + edge engine.

II. VIABILITY: The Plan Is Sustainable

- **Practical Viability**
 - Works purely off phone sensors — no OBD-II or hardware retrofit needed across two-wheelers, trucks and older cars.
- **Scalability**
 - The same trained model scales from a mobile app to FOG-grade IMUs for fleets and defense-grade use.



IMPACT AND BENEFITS

Impact:



Drivers

Continuous nav
No OBD retrofit



Fleets/OEMs

Edge-engine reuse
Predictable ETAs



Year-1 Targets

< 100 m drift/1 km
10 Hz live positioning



Society

Safer 2-wheeler travel
Fewer missed exits

Benefits:

Benefit	Description
Smarter Navigation	Seamless navigation even in tunnels, underground & dense cities.
Cost-effective	No additional hardware required — works on any Android phone.
Scalable	One AI model for both app & edge engine across multiple industries.
Reliable	Real-time, drift-free positioning with UKF fusion & map matching.
Safer & Faster	Reduces wrong turns, delays and improves road safety.



SMART INDIA
HACKATHON
2026



CONCLUSION



SMART INDIA
HACKATHON
2026

- ✓ TrueTrack ensures continuous, accurate and reliable positioning where GNSS fails.
- ✓ Our AI-driven Dead Reckoning + GNSS fusion works on any vehicle, with no extra hardware.
- ✓ From two-wheelers to trucks, fleets to defense — TrueTrack keeps you on the right track.

***Sensing Every Mile,
Even Without GPS***

TrueTrack

SENSING EVERY MILE, EVEN WITHOUT GPS